

Cross Asset Volatility

Gold Waking Up and RV Model Update

- The **CARV Model** indicates a **short bias in short-dated volatility** across **rates and Asia equities**, with **KWEB** and **XLF** as notable exceptions. We conduct a model update for both the CARV and ETF model in this report, and **experiment with EWMA signals as an alternative, replicatable solution** for long-short ranking.
- We are beginning to see a pick up in **gold call open interest** as well as **retail participation** after the gold holding well on the 4000 floor for the past month. We tested an option replacement strategy to capture potential upside, which offers a significantly improved risk profile compared to a long gold position.
 - **Buy GLD Sep-26 415-440 Call Spread for 3.2 (Ref 389.67)**
 - **Buy Dec-26 Worst of Call on {GLD, SPX} for 1%**
- In the volatility space, **GDX-GLD upside volatility spread** continues to deliver attractive carry, with supply and demand flows potentially favoring the trade.
- We have updated our cross-asset volatility monitor. Implied volatility remains **very low in FX, and moderately low in equity with the exception of the Japan-related underlyings across asset classes**.
- FX gamma model signals a mixed outlook, with USD/JPY, JPY cross-vols and MXN vol appearing vulnerable, while EUR, NOK and GBP/CHF vol remain supported. The model's bearish JPY vol signal should be interpreted cautiously, however, as it largely reflects the recent sharp spike in volatility. We think that **delta-hedged 3M USD/JPY atm/25 delta 1*2 (vega notionals) ratio put spreads** are the right instruments for trading the current USD/JPY setup.
- The FX vol surface model favors firmer convexity in HUF and JPY and flatter HUF term structures, though its skew signals remain mixed and offer little directional conviction.

Global Quantitative and Derivatives Strategy

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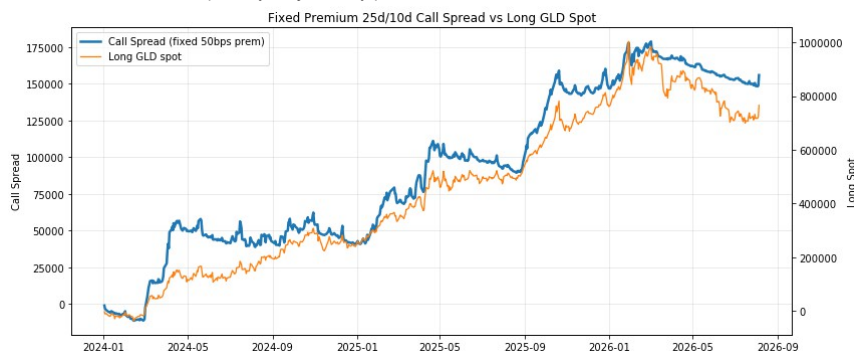
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Figure 1: Call Spread Replacement on GLD vs Long GLD Delta One

\$1M Notional, sized 1/22 per day, Adjusted by premium



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

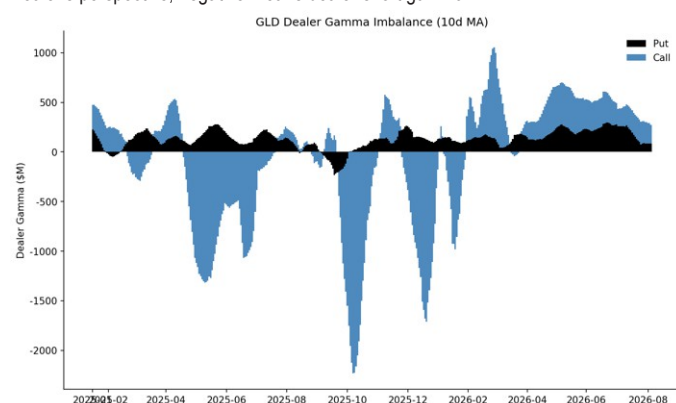
See page 18 for analyst certification and important disclosures.

Gold Waking Up

The near 70% rally in gold from this time last year to Jan-2026 seems a long way away now as the Fed put an end to the party for lucrative **call and call spread buying strategies** ([link](#), [link](#)). As volatility hit new highs, overwriters and underwriters started to dominate the flow, providing more gamma supply (Figure 2), subsequently gold volatility came off the highs but remains elevated by a longer historical measure (Figure 3).

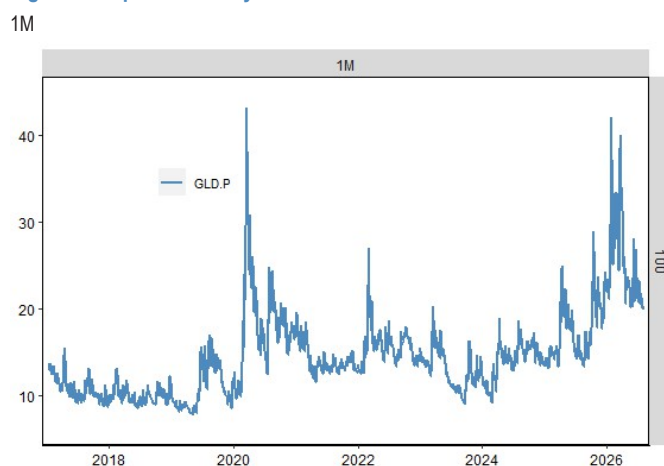
Figure 2: GLD gamma imbalance by put and call

Dealer's perspective, negative means dealer short gamma



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P. OPRA

Figure 3: Implied volatility on GLD

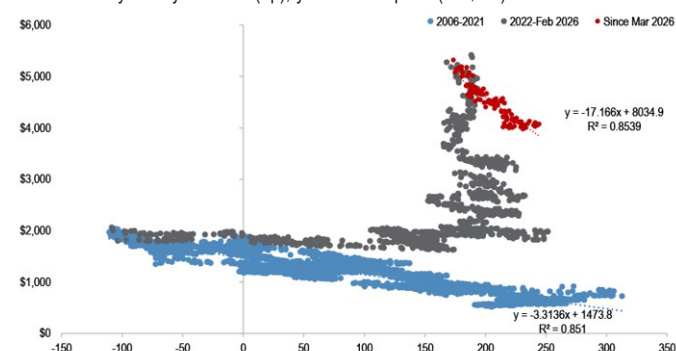


Source: J.P. Morgan Equity Derivatives Strategy

JPM Commodity Research pointed out the near linear relationship with 10Y since March ([link](#)), penalizing the safe haven effect of gold. However, the fundamental view remains on the upside in the long term, as we continue to see strong inflows from central banks with accelerated buying on the dip ([link](#)).

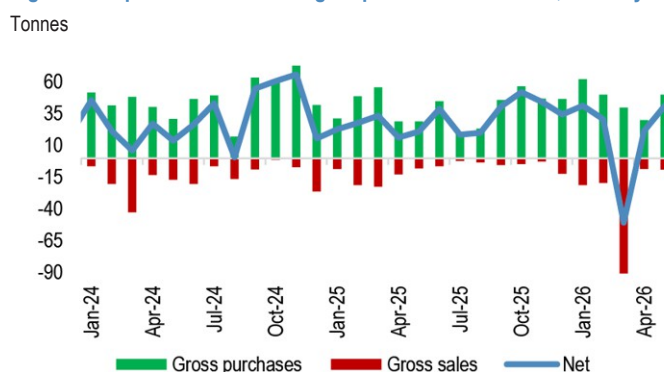
Figure 4: Gold price vs US 10yr real yield level, daily

X-axis: US 10yr real yield level (bp); y-axis: Gold price (US\$/oz)



Source: Bloomberg Finance L.P., J.P. Morgan Commodities Research

Figure 5: Reported central bank gold purchases and sales, monthly



Note: State Oil Fund of the Republic of Azerbaijan (SOFAZ) only reports quarterly data, its reported 1Q26 sales are accounted for in March '26 in this chart.
Source: World Gold Council, IMF, ECB, National Sources, J.P. Morgan Commodities Research

As gold prices are finding the floor at 4000 and trading within a tight 5% range over the whole of July, the first signs of buyers winning over sellers are starting to show with the market digesting news on the inflation pressure.

We are starting to see retail investors warming up to gold again from Wednesday's rally as highlighted in Retail Radar ([link](#)) (Figure 6), as well as a pick up in open interest for GLD calls in the mid-range tenors (Figure 7).

Figure 6: Retail Imbalance in GLD

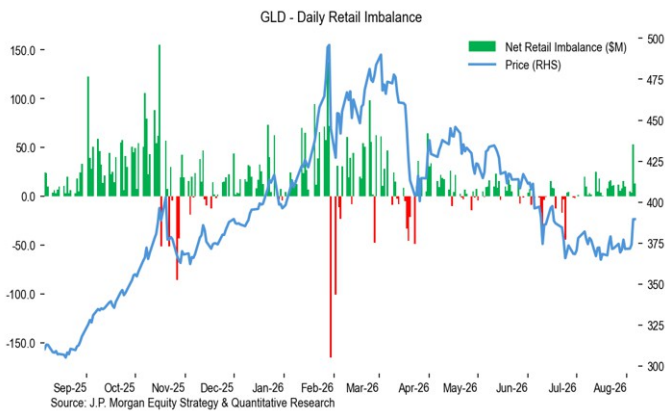
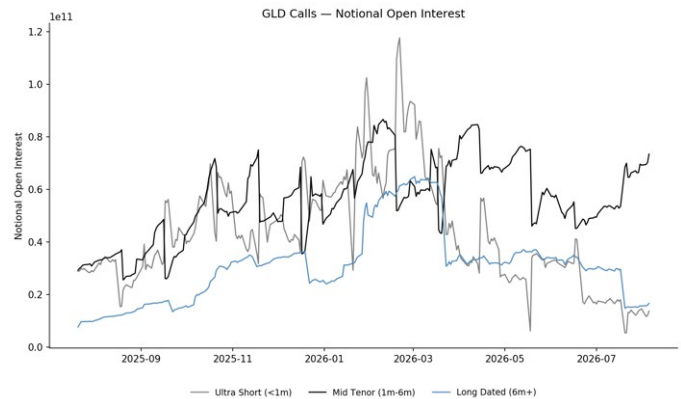


Figure 7: Open Interest Notional Breakdown in GLD Call



With this backdrop, we revisit **cost-effective ways to play upside on gold via liquid options**, taking advantage of the inverted skew (Figure 10). We tested using a **1M 25%-10% delta call spread** on gold compared to **buying gold and holding for one month**, assuming we trade the same cash notional every day held to expiry. If we use a fixed premium approach, the **call spread strategy** delivers a much **higher Sharpe ratio with a much smaller max DD**.

Figure 8: P&L for 1M 25-10 Delta Call Spread on GLD vs Long 1M GLD Spot

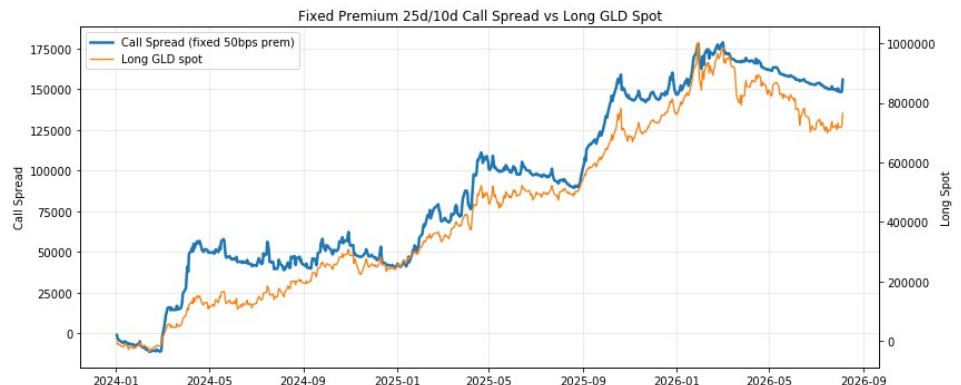
From 2024, No Cost

	Call Spread	Long GLD Spot
Sharpe	1.6	1.36
Max DD (%)	-2.6%	-15.0%

Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

Figure 9: Call Spread Replacement on GLD vs Long GLD Delta One

\$1M Notional, sized 1/22 per day, Adjusted by premium



We can access this from the listed vanilla options with good liquidity. Indicatively,

Buy GLD Sep-26 415-440 Call Spread for 3.2 (Ref 389.67)

(106.5/113% strikes, 0.82% premium, 25/10 delta, max Pnl/Premium 7.8x)

One can also explore using the worst of call with SPX to achieve more leverage with the discount on correlation and low SPX implied volatility.

Buy Dec-26 Worst of Call on {GLD, SPX} for 1% Indicatively

In volatility space, the **spread between GDX and GLD** is getting interesting with good carry. GLD is much **lower vol in absolute values** which makes the underlying the “go to” one for directional plays if momentum start to build up. At current vol levels, **call spread** would likely to be the preferred structured as we suggested. As a result, **higher strikes in GLD will likely have more supply to dealers and reduce the GLD realized volatility if spot does get close**. The GDX flow suggests more **underwriters** and hence more potential volatility delivery for upside call (Figure 33, Figure 35).

This is an ideal set up for long GDX short GLD vol spread, as GDX has a flatter skew, the best entry level is near term upside strikes (Figure 11). Our RV model has also identified this pair as potential long short (Figure 20).

Figure 10: Implied Volatility Surface of GLD

Strike % of forward

	1M	2M	3M	6M	1Y
120	29.9%	27.2%	26.1%	24.5%	23.7%
110	25.0%	24.3%	23.8%	23.2%	22.9%
100	22.3%	22.6%	22.5%	22.5%	22.5%
90	24.2%	23.3%	23.2%	22.8%	22.8%
80	32.8%	29.1%	26.9%	24.7%	23.9%

Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

Figure 11: GDX-GLD Implied Volatility Spread

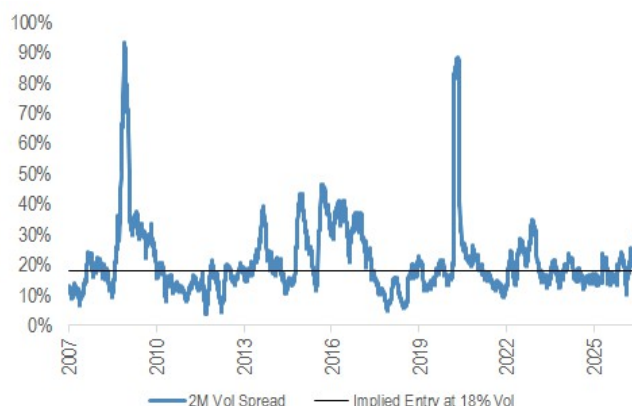
Strike % of forward

	1M	2M	3M	6M	1Y
120	13.9%	17.1%	17.8%	19.3%	19.8%
110	17.2%	18.8%	19.3%	20.0%	20.1%
100	18.9%	19.7%	20.1%	20.3%	20.1%
90	17.6%	19.3%	19.7%	20.0%	19.6%
80	12.1%	15.4%	17.2%	18.4%	18.4%

Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

We favor using **1-2M upside calls for 110% -115% strikes on GDX vs GLD** to capture the potential realized volatility spread.

Figure 12: 2M Realized Volatility Spread of GDX- GLD vs Implied Entry at 18% Vol



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

RV Model Update

We refresh the latest signals from the [CARV Model](#) and the [ETF RV Model](#), and then evaluate the most recent RV performance with the [Cross Asset Volatility RV Strategy](#).

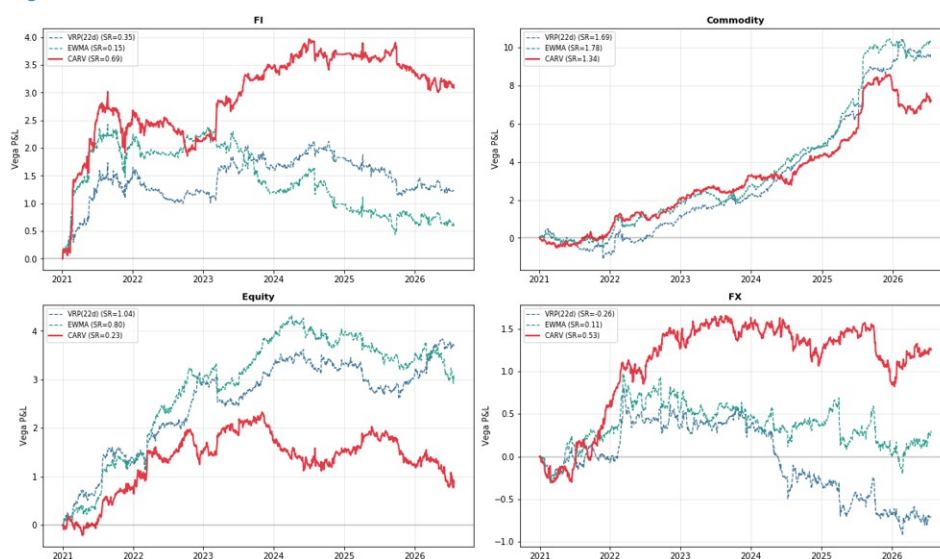
The CARV model indicates a **short bias in short-dated volatility** across **rates and Asia equities**, with **KWEB** and **XLF** being notable exceptions (we have [discussed](#) before owning XLF vol as an effective macro hedge).

Intra-Asset CARV Strategy Performance Update

Given the complexity of the machine-learning implementation, we also experimented with a **simpler, more replicable RV signal** as a benchmark. A common starting point is the volatility risk premium (VRP) - the spread between implied volatility and realized volatility. Volatility clustering motivates forecast-based measures such as EWMA volatility, which places more weight on recent moves. However, EWMA can overweight very large shocks immediately after they occur, so we apply a cap/floor by clipping inputs to dampen outliers. In our setup, we use a 21-day half-life (roughly spanning a 3-month window) and clip at the 2.5th/97.5th percentiles.

We assess signal quality using a stylized cross-sectional RV strategy on 1m variance swaps within each asset class (FI, Commodity, Equity, FX). Each day we rank names by the signal, buy the top basket and sell the bottom basket, using basket sizes of 1/2/3/5 names for FI/Commodity/Equity/FX, and scale positions by inverse volatility of each asset. P&L combines realized-vol accrual and implied-vol mark-to-market, converging at expiry to realized versus implied to mimic var-swap settlement. Note this is not a fully tradable P&L, as it abstracts from hedging path-dependency, higher-order risks, and implementation frictions. Nonetheless, we show a comparison to the dollar P&L of a daily delta-hedged 1M straddle on SMH is broadly consistent (see [Appendix](#)).

Figure 13: Cross Asset Vol RV Backtest

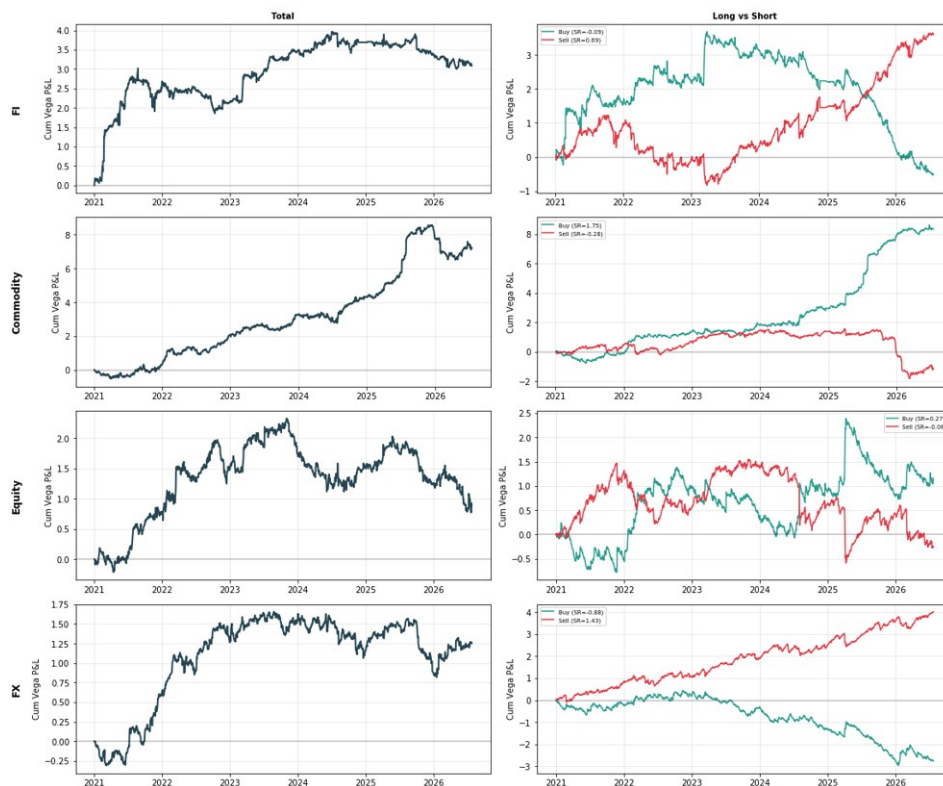


Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P

The CARV model continues to deliver strong outperformance in Fixed Income and FX, demonstrating clear value in asset classes where **volatility exhibits stable mean-reverting dynamics**. FI benefits from a reliable cross-sectional structure, while FX benefits from the model's 2Y rolling-window estimation which adapts well to evolving regimes. From the P&L Split between long/short legs (Figure 14), in **FI and FX** the PnL is mainly driven by **selling expensive vol**, while in **Commodity**, the pnl is driven by **buying cheap vol**, and **Equity is a mixed picture**.

In Equity and Commodity, CARV underperforms relative to simpler benchmarks, though this is largely driven by several **concentrated positions during exceptional regime transitions** - persistent shorts in KOSPI and RTY (Equity) and Silver (Commodity), where these names shifted abruptly to sustained high-vol regimes with limited historical precedent. Notably, both EWMA and CARV suffer during trending vol periods (e.g. KOSPI in 2026), where VRP's shorter lookback allows it to adjust itself faster. This is a known limitation of the model trained on historical patterns: structural breaks by definition fall outside the training distribution.

Figure 14: CARV P&L Split Between Legs



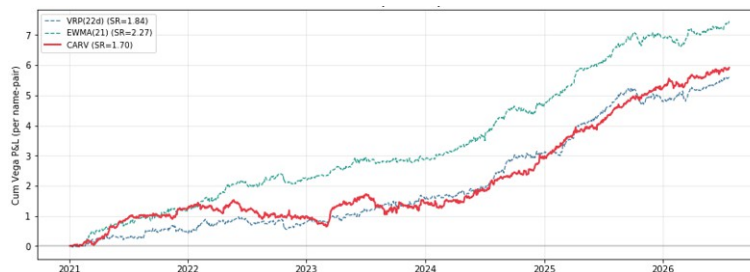
Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P

ETF RV Strategy Performance Update

Separately from signal quality, cross-asset implementation introduces operational friction: option markets differ by asset class in liquidity, contract conventions, and transaction-cost profiles, which can constrain both universe breadth and scalable execution. A practical way to expand breadth is to express cross-asset exposures through ETF options. Our [ETF RV Model](#) utilizes a similar concept, using high frequency data to forecast the “fair” value of implied volatility.

Within the ETF space, we extend our universe to 38 US ETFs ([ETF Universe](#)) with the highest option volume traded and focus on 1M tenor of the volatility curve to ensure the liquidity of our investment universe.

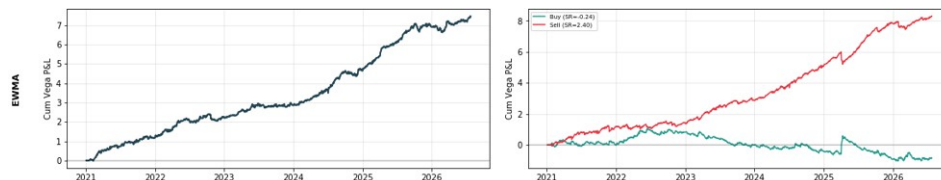
Figure 15: ETF RV backtesting



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

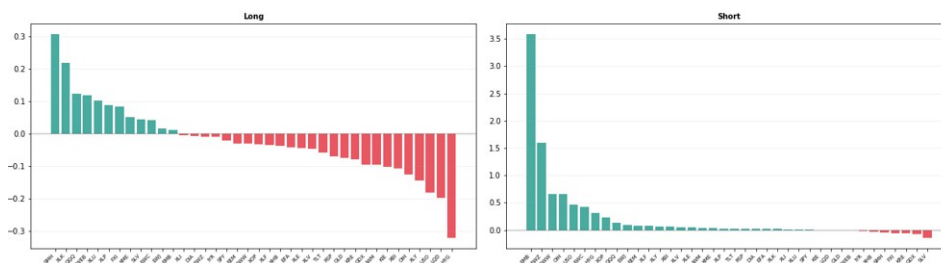
All three strategies deliver strong Sharpe ratios on the ETF universe during the backtest (Figure 15), with PnL overwhelmingly **driven by the sell leg**, and large weights on **selling vol in EM Bond**. The long leg functions as a hedge, where the signal targets the cheapest vol to reduce the bleed. EWMA is the strongest performer, due to the long position in SMH vol in 2022 (Figure 16, Figure 17). This return profile is typically strongest when volatility is stable/mean-reverting, and can be challenged during sharp volatility regime shifts.

Figure 16: EWMA Strategy P&L Split by Long/Short Legs



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

Figure 17: EWMA Strategy P&L Split by Underlyings



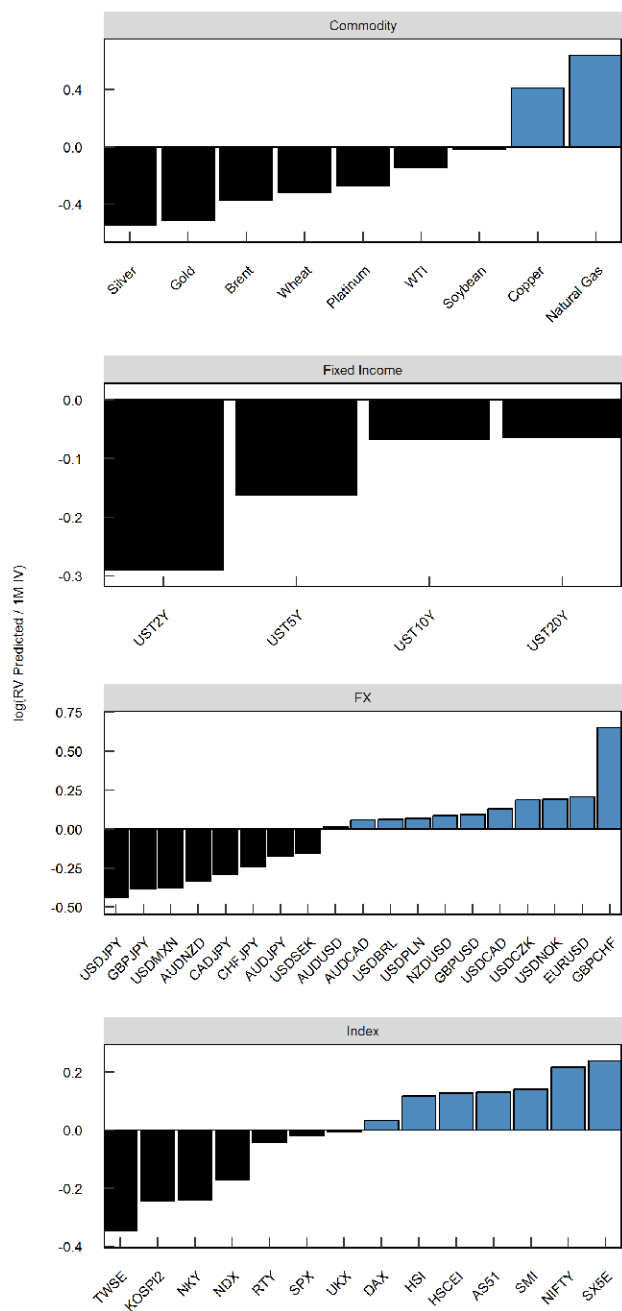
Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

We view the EWMA signal as a simple but effective baseline for relative value volatility positioning. We will continue to enhance the forward realized vol prediction model by incorporating richer features and refining the signal construction to improve the performance.

ML Model Update

Cross Asset Relative Value (CARV) Ranking

Figure 18: Cross Asset Relative Value (CARV) Ranking



Source: J.P. Morgan

Top 15 Richness vol

	Asset	Name	RV Predicted	1M Implied	RV Rank
1	Commodity	Silver	24.9	43.1	1
2	Commodity	Gold	13.6	22.6	2
3	FX	USDJPY	5.6	8.7	3
4	FX	GBPJPY	5.5	8.0	4
5	FX	USDMXN	5.1	7.4	5
6	Commodity	Brent	36.8	53.4	6
7	Index	TWSE	22.8	32.3	7
8	FX	AUDNZD	3.4	4.7	8
9	Commodity	Wheat	24.2	33.2	9
10	FX	CADJPY	6.0	8.1	10
11	FI	UST2Y	1.2	1.6	11
12	Commodity	Platinum	28.4	37.2	12
13	Index	KOSPI2	53.5	68.3	13
14	Index	NKY	22.7	28.8	14
15	FX	CHFJPY	6.5	8.3	15

Source: J.P. Morgan

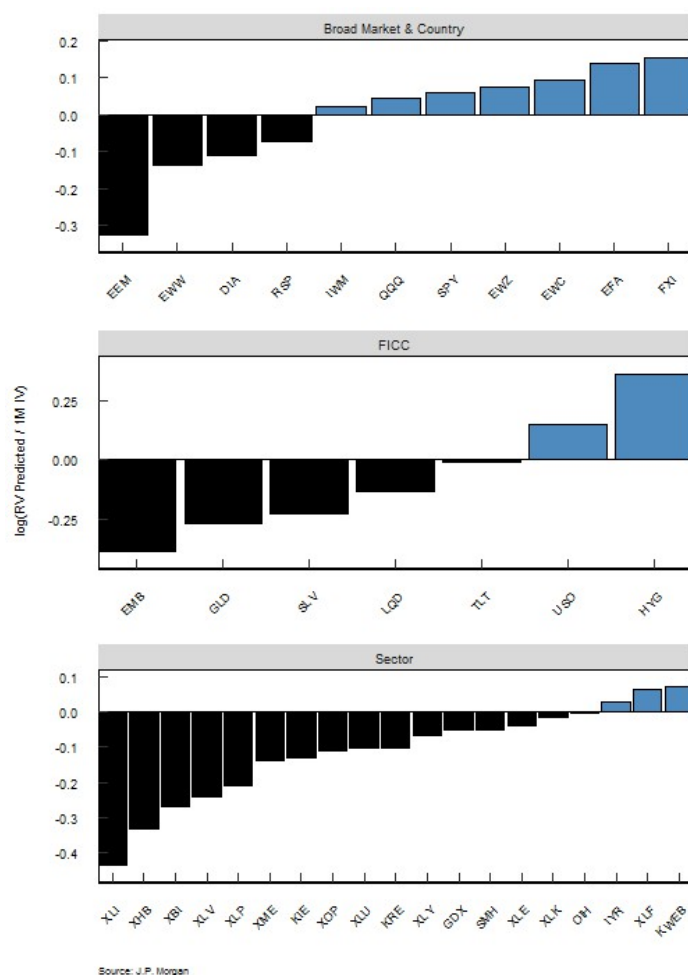
Top 15 Cheapness vol

	Asset	Name	RV Predicted	1M Implied	RV Rank
1	FX	GBPCHF	8.8	4.6	1
2	Commodity	Natural Gas	82.3	43.5	2
3	Commodity	Copper	29.1	19.3	3
4	Index	SX5E	16.6	13.0	4
5	Index	NIFTY	12.8	10.3	5
6	FX	EURUSD	6.2	5.0	6
7	FX	USDNOK	9.9	8.1	7
8	FX	USDCZK	7.1	5.8	8
9	Index	SMI	12.6	10.9	9
10	FX	USDCAD	4.6	4.0	10
11	Index	AS51	12.1	10.6	11
12	Index	HSCEI	21.2	18.6	12
13	Index	HSI	19.3	17.1	13
14	FX	GBPUSD	6.2	5.7	14
15	FX	NZDUSD	8.3	7.7	15

Source: J.P. Morgan

ETF Relative Value Volatility Rankings

Figure 19: ETF Relative Value Volatility Rankings



Top 10 Richness vol

	Name	RV Predicted	1M Implied	RV Rank
1	XLI	12.9	19.9	1
2	EMB	6.1	9.0	2
3	XHB	21.8	30.5	3
4	EEM	21.5	29.8	4
5	XBI	25.6	33.4	5
6	GLD	17.9	23.4	6
7	XLV	13.4	17.1	7
8	SLV	34.9	43.8	8
9	XLP	12.9	15.9	9
10	XME	33.4	38.4	10

Source: J.P. Morgan

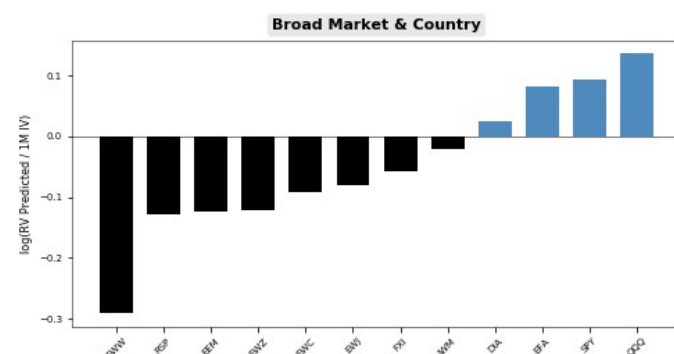
Top 10 Cheapness vol

	Name	RV Predicted	1M Implied	RV Rank
1	HYG	4.7	3.3	1
2	FXI	24.1	20.6	2
3	USO	56.2	48.5	3
4	EFA	16.1	14.0	4
5	EWC	14.7	13.4	5
6	EWZ	27.8	25.8	6
7	KWEB	30.9	28.8	7
8	XLF	15.5	14.6	8
9	SPY	14.0	13.2	9
10	QQQ	22.6	21.6	10

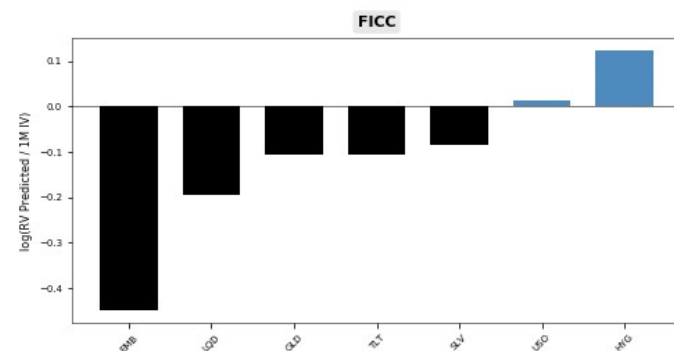
Source: J.P. Morgan

ETF RV Rankings with EWMA Signal

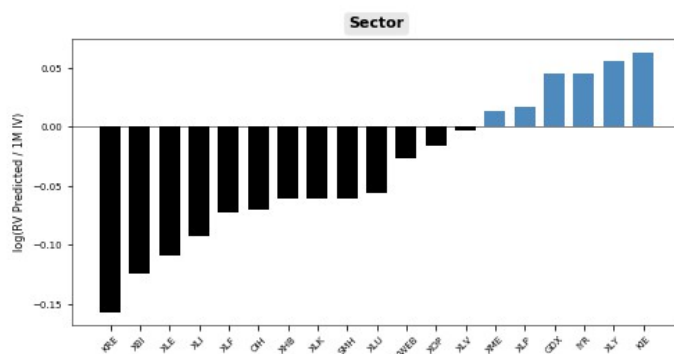
Figure 20: ETF RV Volatility Rankings with EWMA Signal



Name	RV Predicted	1M Implied	RV Rank
EMB	5.7	9.0	1
EWV	19.3	25.8	2
LQD	5.4	6.6	3
KRE	18.9	22.1	4
RSP	11.5	13.0	5
XBI	29.5	33.4	6
EEM	26.3	29.8	7
EWZ	22.8	25.8	8
XLE	22.5	25.1	9
GLD	21.0	23.4	10



Name	RV Predicted	1M Implied	RV Rank
QQQ	24.7	21.6	38
HYG	3.7	3.3	37
SPY	14.5	13.2	36
EFA	15.2	14.0	35
KIE	20.3	19.0	34
XLV	21.0	19.9	33
IVR	15.3	14.6	32
GDX	44.8	42.8	31
DIA	13.7	13.4	30
XLV	16.2	15.9	29

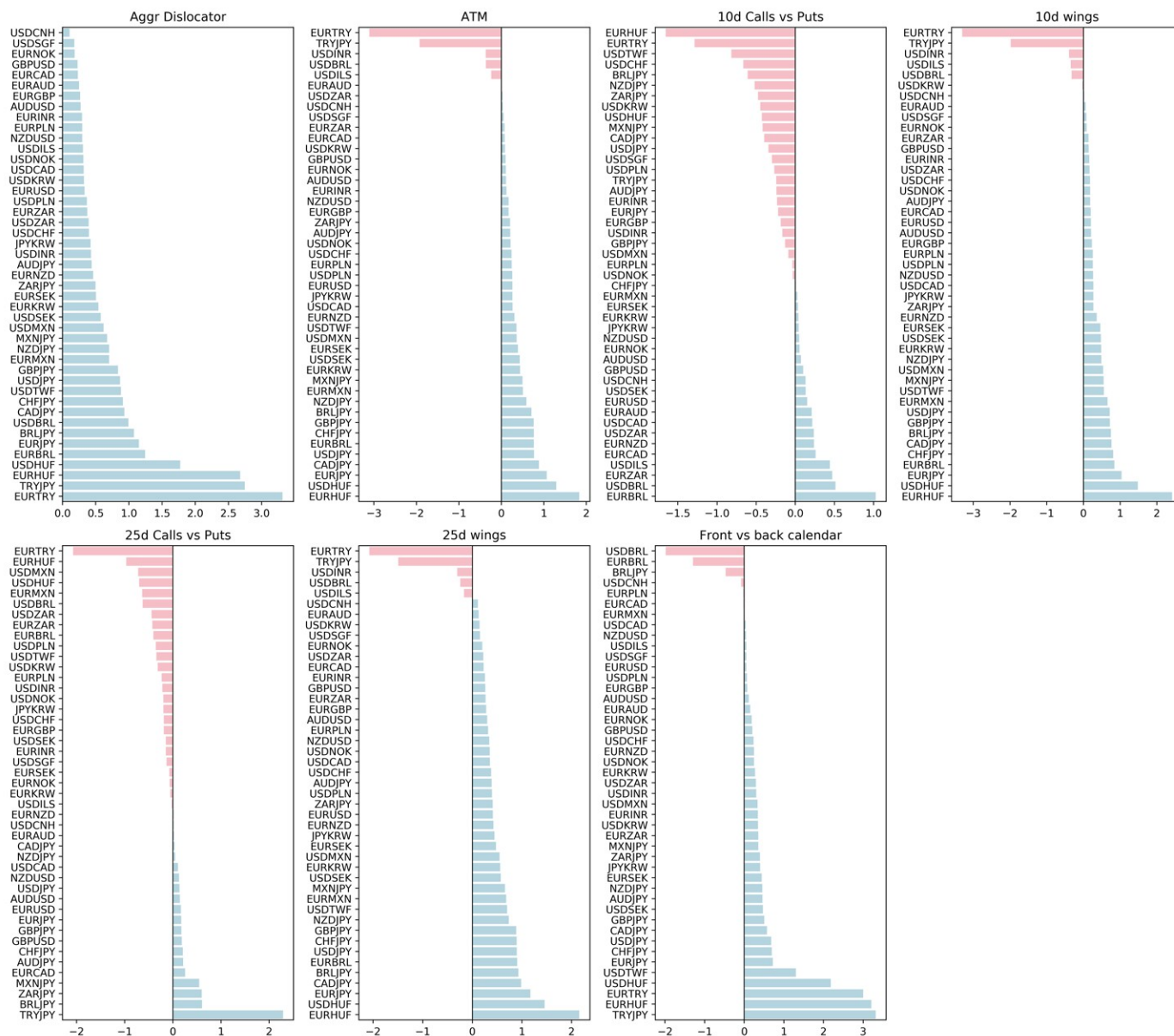


Source: I.P. Morgan

FX Volatility Surface Relative Value

Figure 21: Vol Surface Dislocations Across Various Tradable Axes

Vol Points



Source: J.P. Morgan

Cross Asset Volatility Screen

Figure 22: Parameters of Cross Asset Volatility

		Implied ATM			Realized			Downside Convexity (10dP-50d)			Skew (25dP-25dC)			Upside Skew (50d-25dC)			Term Spread	
		1M	3M	1Y	1M	3M	1Y	1M	3M	1Y	1M	3M	1Y	1M	3M	1Y	3M-1M	1Y-3M
Credit	HYG US Equity	4.0%	5.3%	7.4%	2.8%	3.9%	4.2%	3.0%	4.7%	6.1%	1.9%	2.9%	3.7%	0.7%	1.0%	1.3%	1.3%	2.1%
	LQD US Equity	6.7%	6.7%	8.1%	5.2%	5.5%	5.5%	2.6%	2.9%	3.9%	1.8%	1.8%	2.1%	0.7%	0.6%	0.7%	0.0%	1.3%
Equity	SPX Index	12.5%	14.1%	16.5%	13.8%	13.8%	12.9%	5.6%	8.9%	10.2%	3.1%	4.8%	6.6%	1.0%	1.4%	2.1%	1.6%	2.3%
	NDX Index	20.6%	22.1%	24.3%	25.3%	25.6%	19.6%	5.5%	7.5%	8.6%	3.4%	4.2%	5.1%	1.2%	1.3%	1.6%	1.5%	2.2%
	SXSE Index	12.9%	14.8%	16.8%	14.2%	15.6%	15.5%	4.6%	6.5%	8.4%	2.7%	3.7%	5.1%	0.9%	1.2%	1.8%	1.8%	2.1%
	DAX Index	13.4%	15.1%	16.5%	14.3%	15.9%	16.0%	4.5%	6.5%	8.4%	2.7%	3.8%	5.1%	0.9%	1.2%	1.7%	1.7%	1.4%
	UKX Index	10.6%	11.6%	13.2%	9.7%	11.7%	11.4%	4.4%	5.5%	6.4%	1.9%	2.4%	3.7%	0.5%	0.7%	1.3%	1.1%	1.6%
	SMI Index	10.9%	12.0%	13.5%	9.2%	11.0%	12.1%	4.4%	5.3%	6.1%	2.4%	3.0%	3.7%	0.8%	1.1%	1.4%	1.1%	1.5%
	HSI Index	17.2%	18.6%	20.0%	19.0%	18.5%	18.9%	3.5%	4.0%	4.8%	0.4%	0.8%	1.5%	-0.5%	-0.3%	0.1%	1.4%	1.5%
	NKY Index	28.6%	28.4%	27.0%	35.6%	34.7%	28.3%	7.6%	8.6%	9.1%	3.3%	3.8%	4.2%	0.6%	0.8%	0.9%	-0.3%	-1.3%
FX	EURUSD	5.1%	5.2%	5.9%	4.2%	4.8%	5.6%	0.7%	0.9%	1.3%	0.4%	0.4%	0.4%	0.1%	0.0%	0.0%	0.2%	0.7%
	GBPUSD	5.7%	6.2%	7.2%	6.3%	6.1%	6.4%	0.9%	1.4%	2.2%	0.5%	0.8%	1.3%	0.1%	0.2%	0.3%	0.5%	1.0%
	EURJPY	8.1%	7.8%	7.7%	8.8%	6.2%	6.3%	3.5%	3.0%	1.6%	2.4%	1.8%	0.6%	0.8%	0.5%	0.0%	-0.4%	0.0%
	USDJPY	8.7%	8.2%	8.2%	9.9%	6.8%	8.0%	3.4%	2.8%	1.3%	2.5%	1.7%	0.2%	0.9%	0.5%	-0.2%	-0.5%	-0.1%
	AUDUSD	7.2%	7.4%	8.5%	6.2%	7.2%	8.0%	1.2%	1.5%	2.1%	0.6%	0.8%	1.0%	0.1%	0.2%	0.2%	0.3%	1.1%
	CADUSD	4.1%	4.2%	4.9%	3.8%	3.5%	4.3%	0.3%	0.4%	0.7%	-0.2%	-0.1%	-0.1%	-0.2%	-0.2%	-0.2%	0.1%	0.7%
Commod	XAU	22.6%	22.3%	21.5%	23.9%	25.1%	27.5%	1.3%	2.1%	2.8%	-1.0%	-0.8%	-0.5%	-1.1%	-1.2%	-1.2%	-0.3%	-0.8%
	WTI	52.6%	44.5%	29.5%	65.4%	57.4%	54.0%	-3.1%	-3.5%	5.0%	-9.8%	-12.2%	0.8%	-6.6%	-8.5%	-0.7%	-8.1%	-15.0%
Rate	USD_2Y	5.4	5.7	6.1	1.3	1.3	1.2	0.1	0.0	0.1	-0.3	-0.2	-0.3	0.8	0.8	0.8	0.3	0.4
	USD_5Y	5.1	5.3	5.6	2.9	3.1	2.7	0.1	0.2	0.2	-0.3	-0.3	-0.3	0.6	0.4	0.5	0.2	0.3
	USD_10Y	4.5	4.6	5.1	4.3	4.6	4.1	0.2	0.3	0.3	-0.3	-0.3	-0.3	0.3	0.3	0.3	0.2	0.5
	USD_30Y	4.9	5.3	5.6	7.8	8.4	7.6	0.8	0.7	0.7	-0.6	-0.4	-0.4	-0.2	-0.3	-0.2	0.4	0.3
	EUR_2Y	3.7	3.9	4.4	1.2	1.3	1.2	0.1	0.2	0.7	-0.1	-0.2	-0.3	0.0	0.0	-0.4	0.3	0.4
	EUR_10Y	4.5	4.7	4.9	4.4	4.9	4.6	0.9	0.8	0.7	-0.7	-0.5	-0.5	-0.2	-0.1	0.0	0.2	0.3
	GBP_10Y	4.0	4.2	4.6	9.7	10.0	9.4	0.7	0.7	0.7	-0.7	-0.5	-0.4	-0.1	-0.1	-0.1	0.3	0.4
	JPY_10Y	2.9	3.2	3.8	4.0	4.6	3.7	0.0	0.1	0.1	-0.4	-0.2	-0.2	0.5	0.4	0.5	0.3	0.7

*Rate implied volatility are from swaption and realized are based on bond futures, convention bps
Source: J.P.Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

Figure 23: 2Y %tile of Cross Asset Volatility Parameters

		Implied ATM			Realized			Downside Convexity (10dP-50d)			Skew (25dP-25dC)			Upside Skew (50d-25dC)			Term Spread	
		1M	3M	1Y	1M	3M	1Y	1M	3M	1Y	1M	3M	1Y	1M	3M	1Y	3M-1M	1Y-3M
Credit	HYG US Equity	16.8%	44.9%	66.0%	8.6%	22.7%	0.6%	16.8%	62.0%	52.0%	20.3%	68.6%	64.8%	23.5%	69.0%	66.5%	92.4%	86.0%
	LQD US Equity	47.0%	27.7%	13.6%	31.3%	27.7%	1.2%	79.3%	66.7%	61.0%	85.9%	82.2%	75.7%	90.8%	89.3%	80.9%	16.1%	52.0%
Equity	SPX Index	31.4%	39.2%	56.4%	61.8%	54.6%	39.4%	5.5%	20.8%	39.0%	3.6%	6.1%	10.1%	4.6%	5.4%	5.7%	80.5%	92.0%
	NDX Index	68.1%	77.1%	89.9%	82.3%	86.1%	47.6%	2.5%	3.3%	24.1%	3.4%	1.3%	1.5%	5.5%	1.1%	1.7%	66.7%	93.7%
	SXSE Index	19.5%	40.3%	71.5%	53.2%	58.7%	41.5%	3.6%	0.8%	35.9%	4.6%	3.4%	41.5%	6.7%	6.7%	50.3%	96.0%	97.9%
	DAX Index	31.7%	49.7%	68.6%	53.1%	63.3%	41.8%	3.1%	5.4%	37.3%	7.8%	9.4%	46.5%	18.0%	16.8%	48.2%	93.5%	92.7%
	UKX Index	52.2%	59.5%	67.3%	54.4%	66.3%	48.5%	34.4%	29.4%	49.5%	23.3%	9.8%	50.9%	17.0%	7.3%	59.1%	74.2%	73.2%
	SMI Index	20.1%	31.7%	52.0%	14.0%	35.8%	41.4%	18.4%	16.8%	40.0%	19.9%	25.0%	56.4%	27.2%	48.9%	68.5%	90.6%	95.2%
	HSI Index	11.5%	7.3%	5.0%	49.7%	32.0%	12.6%	62.7%	65.2%	77.6%	58.7%	69.6%	82.0%	57.9%	73.4%	82.2%	69.0%	72.8%
	NKY Index	81.5%	85.9%	88.1%	82.8%	86.5%	81.8%	37.5%	42.3%	65.6%	19.9%	17.6%	36.5%	7.8%	1.0%	1.9%	48.0%	36.5%
FX	EURUSD	5.8%	3.5%	0.4%	14.2%	10.1%	0.9%	60.0%	50.4%	55.2%	65.6%	66.5%	64.4%	69.1%	68.1%	64.8%	68.0%	82.6%
	GBPUSD	4.6%	5.9%	4.4%	36.6%	22.8%	23.0%	47.2%	55.4%	64.4%	50.9%	63.1%	78.1%	56.3%	65.4%	80.7%	86.1%	93.3%
	EURJPY	45.2%	26.7%	10.7%	69.1%	23.8%	15.7%	98.3%	95.4%	7.2%	96.7%	89.3%	10.4%	95.4%	79.3%	27.8%	14.3%	30.7%
	USDJPY	25.6%	13.8%	1.3%	59.5%	6.6%	3.3%	95.6%	89.7%	6.1%	95.0%	86.7%	8.7%	95.2%	75.7%	14.0%	24.3%	49.5%
	AUDUSD	3.9%	1.7%	0.2%	16.8%	19.5%	2.0%	27.4%	11.9%	12.6%	27.8%	25.0%	25.0%	34.3%	33.1%	42.6%	56.7%	90.7%
	CADUSD	2.6%	0.4%	3.1%	24.3%	0.7%	2.4%	41.4%	45.9%	64.3%	49.4%	58.0%	62.2%	50.1%	59.7%	66.1%	52.7%	81.8%
Commod	XAU	81.9%	81.5%	75.6%	72.6%	81.6%	100.0%	74.4%	81.9%	89.1%	34.5%	77.8%	91.5%	18.4%	46.1%	91.9%	36.1%	20.2%
	WTI	79.3%	78.1%	50.2%	90.5%	83.0%	99.8%	15.7%	15.1%	2.2%	17.3%	13.2%	8.6%	18.3%	14.0%	14.4%	19.6%	19.3%
Rate	USD_2Y	32.4%	37.4%	40.2%	32.6%	27.3%	0.4%	10.8%	3.5%	23.8%	93.5%	93.1%	95.8%	8.3%	7.3%	5.2%	68.8%	62.8%
	USD_5Y	32.4%	33.5%	39.1%	39.5%	34.5%	0.4%	16.0%	6.3%	27.9%	96.3%	93.7%	96.5%	4.2%	4.0%	3.7%	52.5%	73.0%
	USD_10Y	25.3%	21.8%	30.3%	33.6%	37.2%	0.4%	8.7%	8.8%	15.4%	97.9%	96.5%	80.2%	2.3%	5.8%	23.1%	47.3%	88.7%
	USD_30Y	23.0%	18.2%	21.6%	32.4%	34.4%	0.8%	0.6%	60.2%	58.5%	90.2%	92.3%	37.5%	7.1%	7.9%	51.9%	36.2%	91.2%
	EUR_2Y	58.7%	58.4%	57.1%	53.5%	38.3%	38.3%	4.3%	6.4%	9.4%	89.6%	92.6%	90.8%	12.2%	11.3%	9.2%	58.4%	31.5%
	EUR_10Y	23.6%	31.0%	40.5%	29.2%	38.0%	12.0%	14.5%	10.9%	11.3%	89.2%	92.1%	96.0%	7.2%	5.1%	4.0%	81.2%	91.9%
	GBP_10Y	39.0%	36.5%	35.4%	74.3%	68.3%	89.2%	3.2%	2.5%	4.6%	95.4%	97.3%	95.4%	4.2%	4.9%	6.3%	63.7%	49.1%
	JPY_10Y	70.9%	85.6%	94.8%	80.3%	85.8%	58.0%	43.1%	70.2%	5.8%	67.3%	66.3%	95.8%	30.8%	19.6%	3.8%	94.8%	99.3%

Source: J.P.Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

Appendix - P&L of Long Straddle vs Varswap

Figure 24: SMH 1M Straddle Dollar PnL vs Volatility Carry

1M notional on straddle; 1 vega notional in varswap



Source: J.P.Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

Appendix - ETF RV Strategy Universe

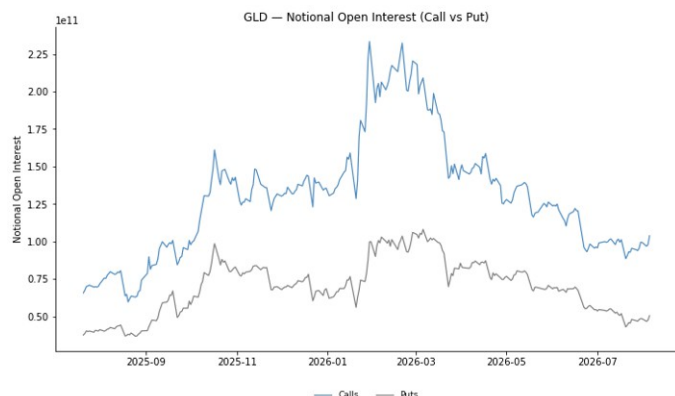
Figure 25: ETF RV Strategy Universe

SPY	FXI	XLV	SLV
QQQ	XLE	XLY	EMB
IWM	EWZ	OIH	XLI
DIA	XLU	KWEB	XHB
EEM	XBI	USO	KIE
EFA	IYR	GLD	EWG
SMH	KRE	TLT	EWJ
XLF	XLK	HYG	EWV
XOP	GDX	LQD	RSP
XLP	XME		

Appendix - Flow in Gold Options

Figure 26: Open Interest Notional in GLD Options

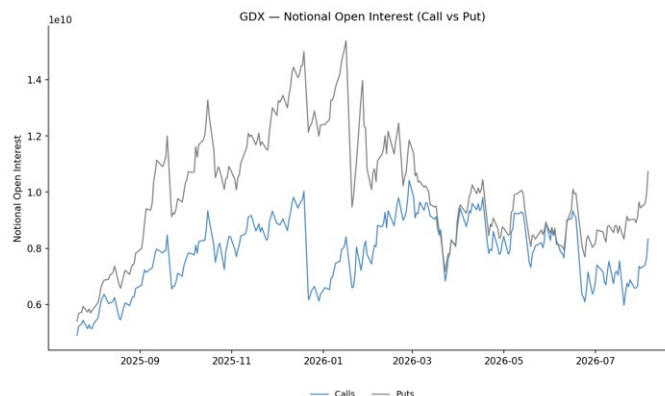
As of August 5th



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P. OPRA

Figure 27: Open Interest Notional in GDX Options

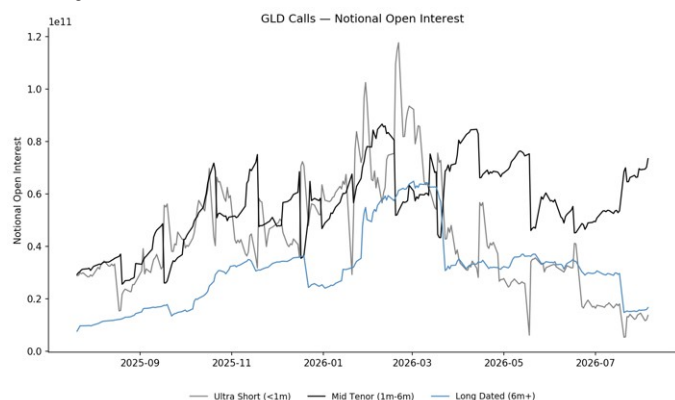
As of August 5th



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P. OPRA

Figure 28: Open Interest Notional Breakdown in GLD Call

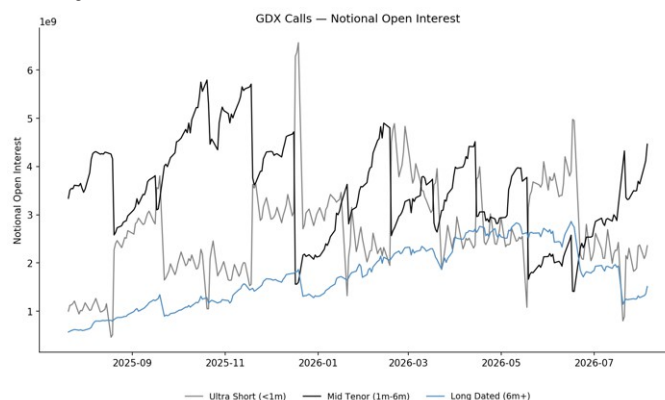
As of August 5th



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P. OPRA

Figure 29: Open Interest Notional Breakdown in GDX Call

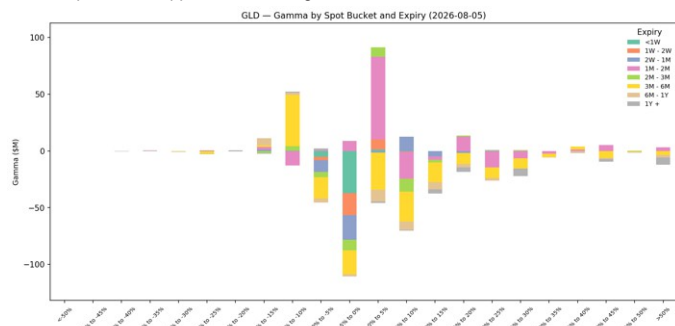
As of August 5th



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P. OPR

Figure 30: Gamma Imbalance from Listed Options on GLD

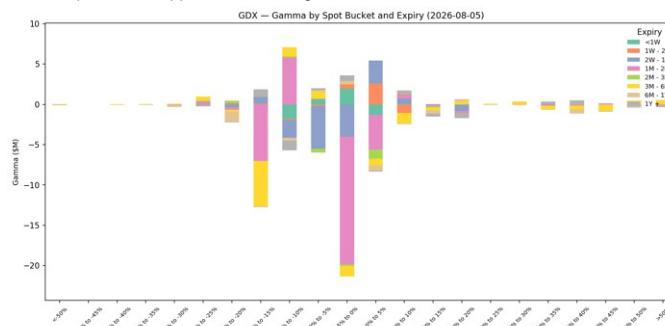
Dealer's position is opposite, As of August 5th



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P. OPRA

Figure 31: Gamma Imbalance from Listed Options on GDX

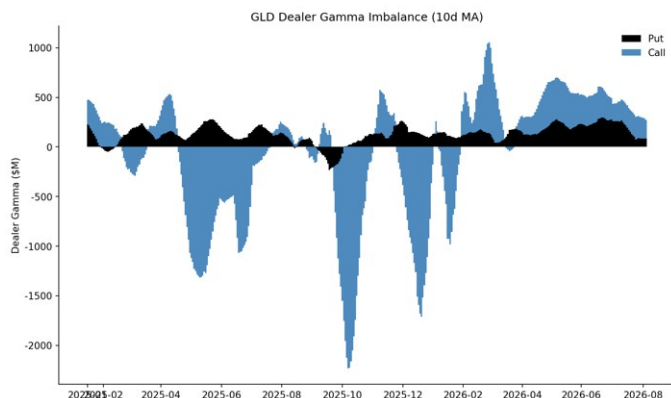
Dealer's position is opposite, As of August 5th



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P. OPRA

Figure 32: GLD Gamma Imbalance by Put and Call

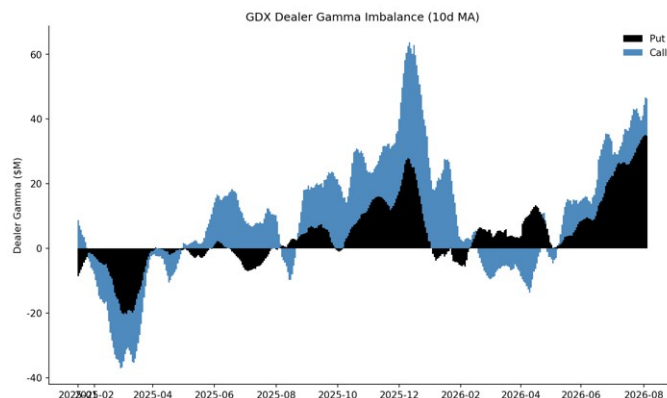
10-Day Moving Average, as of August 5th



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P. OPRA

Figure 33: GDX Gamma Imbalance by Put and Call

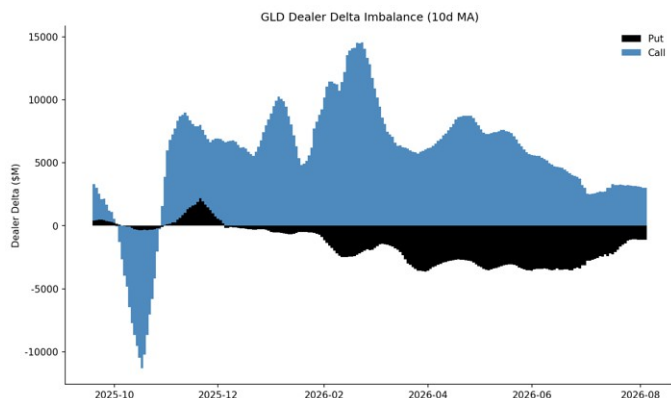
10-Day Moving Average, as of August 5th



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P. OPRA

Figure 34: GLD Delta Imbalance by Put and Call

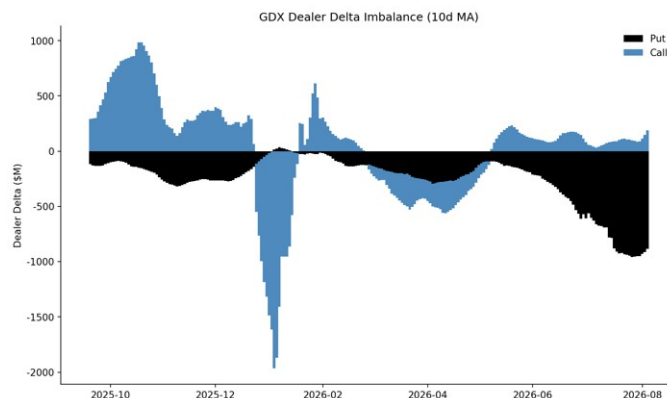
10-Day Moving Average, as of August 5th



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P. OPRA

Figure 35: GDX Delta Imbalance by Put and Call

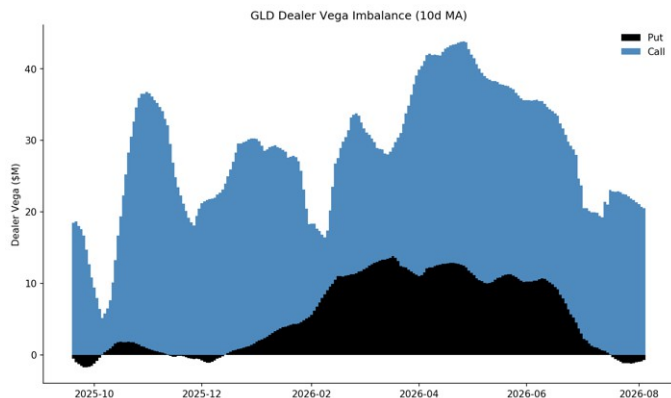
10-Day Moving Average, as of August 5th



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P. OPRA

Figure 36: GLD Vega Imbalance by Put and Call

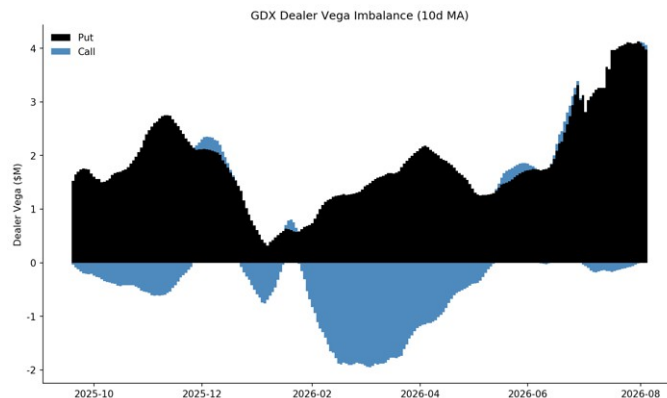
10-Day Moving Average, as of August 5th



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P. OPRA

Figure 37: GDX Vega Imbalance by Put and Call

10-Day Moving Average, as of August 5th



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P. OPRA

Appendix - Related Publications

Cross Asset Volatility Strategy

- [Cross Asset Volatility: What to Long, Short, and When - \(Commodity Part I\)](#)
- [Cross Asset Volatility: 2022 Deep Dive, Oil and Gold Update, FOMC and ML Model Update](#)
- [Cross Asset Volatility: Oil Derivatives and “Cheaper To Deliver” Macro Hedges](#)
- [Cross Asset Volatility: Gold Through the Lens of Derivatives: Positioning and Trade](#)
- [Cross Asset Volatility: Equity-USD, Buy the Dip, Japan and ML Model Update](#)
- [Cross Asset Volatility: FOMC Implied Moves, AI Credit Hedge and ML Model Output Update](#)
- [Cross Asset Volatility: Gold ETF Option flow, FTSE vs Sterling and Machine Learning Model Output Update](#)
- [Cross Asset Volatility: Record Short Gamma in Gold ETF and Machine Learning Based Trade Recommendations](#)

Equity Volatility Strategy

- [Equity Volatility Strategy: Stellar P&L from Earnings Dispersion Echoes of 1998, How to Buy Correlation?](#)
- [Equity Volatility Strategy: Intraday Momentum Deep Dive \(Part I\)](#)
- [Equity Volatility Strategy: Have Parameters Normalized Too Much?](#)
- [Equity Volatility Strategy: What To Long, Short And When \(Part II\)](#)
- [Equity Volatility Strategy: “Too Big to Move”? Dispersion Update and RV Opportunities](#)
- [Equity Volatility Strategy: Lesson From The 2022 Playbook](#)
- [Equity Volatility Strategy: Vol RV opportunities amid the software correction](#)
- [Equity Volatility Strategy: What to Long, Short, and When \(Part I\)](#)
- [Equity Volatility Strategy: Why index vol is so low and MXWO dispersion](#)
- [Equity Volatility Strategy: Term-structure inversion, NVDA, Gamma dispersion and VIX](#)
- [Equity Volatility Strategy: Dispersion Update, AI vs Dot-Com and Earnings](#)
- [Equity Volatility Strategy: Take some chips off the table, replace with leveraged options](#)
- [Equity Volatility Strategy Thinking Outside the Box for Tail Trading](#)
- [Equity Volatility Strategy Thinking Outside the Box for Tail Trading - Part 2](#)

FX Derivatives Strategy

- [FX Derivatives: A shade more cautious](#), April 24

- [JPM FX - Derivatives Chartpack Notes: USD skew normalization continues - fade in USD/MXN and USD/INR](#), April 17
- [FXO Quant: To Cease Or Not To Cease](#), April 17
- [FX Derivatives: Steering Through the Endgame](#), April 10
- [JPM FX - Derivatives Chartpack Notes: Vols on a Roll. What's lagging the repricing moves?](#), April 8
- [FX Derivatives: Kicking The Can](#), March 27

Risks of Common Option Strategies

Risks to Strategies: Not all option strategies are suitable for investors; certain strategies may expose investors to significant potential losses. We have summarized the risks of selected derivative strategies. For additional risk information, please call your sales representative for a copy of “Characteristics and Risks of Standardized Options.” We advise investors to consult their tax advisors and legal counsel about the tax implications of these strategies. Please also refer to option risk disclosure documents.

Put Sale: Investors who sell put options will own the underlying asset if the asset’s price falls below the strike price of the put option. Investors, therefore, will be exposed to any decline in the underlying asset’s price below the strike potentially to zero, and they will not participate in any price appreciation in the underlying asset if the option expires unexercised.

Call Sale: Investors who sell uncovered call options have exposure on the upside that is theoretically unlimited.

Call Overwrite or Buywrite: Investors who sell call options against a long position in the underlying asset give up any appreciation in the underlying asset’s price above the strike price of the call option, and they remain exposed to the downside of the underlying asset in the return for the receipt of the option premium.

Booster : In a sell-off, the maximum realized downside potential of a double-up booster is the net premium paid. In a rally, option losses are potentially unlimited as the investor is net short a call. When overlaid onto a long position in the underlying asset, upside losses are capped (as for a covered call), but downside losses are not.

Collar: Locks in the amount that can be realized at maturity to a range defined by the put and call strike. If the collar is not costless, investors risk losing 100% of the premium paid. Since investors are selling a call option, they give up any price appreciation in the underlying asset above the strike price of the call option.

Call Purchase: Options are a decaying asset, and investors risk losing 100% of the premium paid if the underlying asset’s price is below the strike price of the call option.

Put Purchase: Options are a decaying asset, and investors risk losing 100% of the premium paid if the underlying asset’s price is above the strike price of the put option.

Straddle or Strangle: The seller of a straddle or strangle is exposed to increases in the underlying asset’s price above the call strike and declines in the underlying asset’s price below the put strike. Since exposure on the upside is theoretically unlimited, investors who also own the underlying asset would have limited losses should the underlying asset rally. Covered writers are exposed to declines in the underlying asset position as well as any additional exposure should the underlying asset decline below the strike price of the put option. Having sold a covered call option, the investor gives up all appreciation in the underlying asset above the strike price of the call option.

Put Spread: The buyer of a put spread risks losing 100% of the premium paid. The buyer of higher-ratio put spread has unlimited downside below the lower strike (down to zero), dependent on the number of lower-struck puts sold. The maximum gain is limited to the spread between the two put strikes, when the underlying is at the lower strike. Investors who own the underlying asset will have downside protection between the higher-strike put and the lower-strike put. However, should the underlying asset’s price fall below the strike price of the lower-strike put, investors regain exposure to the underlying asset, and this exposure is multiplied by the number of puts sold.

Call Spread: The buyer risks losing 100% of the premium paid. The gain is limited to the spread between the two strike prices. The seller of a call spread risks losing an amount equal to the spread between the two call strikes less the net premium received. By selling a covered call spread, the investor remains exposed to the downside of the underlying asset and gives up the spread between the two call strikes should the underlying asset rally.

Butterfly Spread: A butterfly spread consists of two spreads established simultaneously – one a bull spread and the other a bear spread. The resulting position is neutral, that is, the investor will profit if the underlying is stable. Butterfly spreads are established at a net debit. The maximum profit will occur at the middle strike price; the maximum loss is the net debit.

Pricing Is Illustrative Only: Prices quoted in the above trade ideas are our estimate of current market levels, and are not indicative trading levels.

Disclosures

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